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Title

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Abstract

The inherently low thermal conductivity of conventional hole-transport layers (HTLs) in inverted perovskite solar cells (PSCs) introduces a substantial discrepancy in interlayer heat-transfer dynamics, leading to detrimental heat accumulation and nonradiative recombination. Herein, we develop a spinel-type semiconductor of CuBi_2O_4 , and integrate it into a composite HTL architecture to regulate heat conduction for the first time. Leveraging enhanced phonon group velocities, the CuBi_2O_4 -based composite HTL achieves exceptional thermal compatibility with the perovskite absorber, demonstrating enhanced heat conduction and optimal thermal-expansion coefficient alignment. These synergistic effects significantly delay hot-carrier relaxation and reduce excess energy dissipation by approximately 10-fold. Consequently, we obtain high-quality perovskite films with ordered orientation and released residual strain, yielding an impressive power conversion efficiency (PCE) of 27.18% (certified 26.83%). Remarkably, these phonon-engineered devices maintain 90.1%, 82.3%, 85.6% and 93.7% of their initial PCEs under ISOS-D-2I, ISOS-D-3, ISOS-T-1 and ISOS-L-1 conditions for 2000 h, respectively.

Keywords: CuBi_2O_4 nanocrystals; Composite hole-transport layer; Interface heat conduction; Inverted perovskite solar cells; Device stability

Introduction

The last decade has witnessed an astonishing research improvement of perovskite solar cells (PSCs) with a certified power conversion efficiency over 27%, a progress mainly driven by materials innovation and device engineering¹⁻⁹. However, the heat generated during the persistent operation process of PSCs and originating from the bulk and air environment inevitably poses significant challenges to the power conversion efficiency (PCE) and operational stability of PSCs as they move towards commercialization¹⁰. The generation of heat during the operation of PSCs mainly originates from (1) energy dissipation caused by thermal relaxation and nonradiative recombination, and (2) Joule heat at reverse bias¹¹. The operational temperature of PSCs is over 70 °C under practical application conditions. These thermal stresses lead to severe ion migration, lattice expansion, hotspot formation and heat accumulation, which are detrimental to the efficiency and stability of PSCs¹². To date, the heat dissipation has been a critical concern topic. Previous strategies, such as improving the thermal conductivity (κ) of perovskite films by crystallographic orientation and halide alloying have been regarded as an efficient strategy^{12,13}. Moreover, inserting a heat dissipation layer is also a feasible method to transfer heat and reduce the surface temperature^{14,15}. However, these approaches primarily address heat conduction to a limited extent and only within the bulk perovskite or at the isolated interface, neglecting the critical thermal compatibility between perovskites and intrinsic charge-transport materials—the primary bottleneck for heat dissipation.

Unfortunately, most popular organic hole transport materials (HTMs) (for example, poly(3-hexylthiophene-2,5-diyl) (P3HT), (4-(3,6-dimethyl-9H-carbazol-9-yl)butyl)phosphonic acid (Me-4PACz)) all exhibit extremely low thermal conductivities of $< 0.5 \text{ W m}^{-1} \text{ K}^{-1}$, whereas the thermal conductivity of perovskites is $> 1 \text{ W m}^{-1} \text{ K}^{-1}$ ¹⁶. It is difficult to realize exceptional thermal compatibility by altering perovskite component or introducing additives, owing to the fixed chemical composition of high-efficiency perovskites. Meanwhile, these outstanding photoelectric properties of organic HTMs are irreplaceable in high-efficiency devices, but their extremely low thermal conductivity greatly limit heat conduction at the buried interface. In contrast, inorganic HTMs feature a high thermal conductivity of $0.5\sim 5 \text{ W m}^{-1} \text{ K}^{-1}$ and variable chemical composition¹⁷. Given that poor thermal compatibility not only affects heat dissipation, but also influences perovskite crystallization and charge-carrier dynamics. The huge differences in thermal conductivity between the HTL and perovskite will cause significant interfacial phonon scattering, which instead increase the overall thermal resistance and limit heat dissipation¹⁸. Additionally, overly fast interfacial heat transfer can affect the relaxation of hot carriers through phonon coupling, triggering more nonradiative recombination¹⁹. Meanwhile, overly slow interfacial heat transfer can induce localized undercooling or overheating of perovskite films, thereby leading to polycrystallization and grain-boundary defects²⁰. So far, no attempts have succeeded in achieving suitable thermal compatibility between perovskites and HTMs, including all reported inorganic HTMs. Therefore, it is urgent to tailor interlayer heat-transfer dynamics. Fortunately, we find that the organic-inorganic composite hole-transport layer (HTL) has a huge potential to precisely improve thermal compatibility with perovskites.

In this contribution, we establish a feasible phonon-engineered strategy for advanced composite HTLs, addressing a long-standing bottleneck of inefficient heat conduction in inverted PSCs. Specifically, we develop a spinel-type semiconductor, CuBi_2O_4 , and employ it to construct a composite HTL. This composite HTL feature a high thermal conductivity derived from enhanced phonon group velocities, thus achieving enhanced heat conduction and optimal thermal-expansion coefficient alignment. Therefore, our champion inverted PSCs yield remarkable PCEs of 27.18% (0.09 cm^2 , certified 26.83%), 25.75% (1 cm^2), 19.01% (210 cm^2) and 44.07% (indoor, 1000 lux), respectively. Moreover, the CuBi_2O_4 -based PSCs exhibit exceptional operational stability under ISOS-D-2I, ISOS-D-3, ISOS-T-1 and ISOS-L-1 conditions.

Results

Preparations of CuBi_2O_4 -based composite HTLs.

In view of the differences in thermal conductivity between each layer, we first measured thermal constant analyzer (TCA). Fig. 1a and Supplementary Fig. 1 showed the thermal conductivity of the (9H,9'H-(3,3'-bicarbazole)-9,9'-diylbis(butane-4,1-diyl))diphosphonic acid (DCZ-4P) film was only $0.186 \text{ W m}^{-1} \text{ K}^{-1}$ at 300 K (Supplementary Table 1), while that of the perovskite film was $1.168 \text{ W m}^{-1} \text{ K}^{-1}$ (Supplementary Table 2). Even if the temperature rose, the thermal conductivity of DCZ-4P did not show any significant change. A large difference in thermal conductivity was observed between DCZ-4P and the perovskite layer. In contrast, the difference in thermal conductivity between electronic transport layers and perovskite films was negligible. These results suggested that the HTL/buried-perovskite region might be one of the important bottlenecks for heat transfer in inverted PSCs. The corresponding thermal conductivity of traditional HTMs (NiO_x)

was also determined to be $0.78 \text{ W m}^{-1} \text{ K}^{-1}$ at 300 K by TCA measurement. It showed a poor thermal compatibility (Supplementary Fig. 2), even when a composite layer was constructed by chelating DCZ-4P and NiO_x . Fortunately, spinel-type oxides (e.g. NiCo_2O_4 , CuCo_2O_4) exhibited high thermal conductivity ($> 1 \text{ W m}^{-1} \text{ K}^{-1}$) and tunable electrical properties^{21,22}. Other HTMs with high thermal conductivity, such as graphene ($> 100 \text{ W m}^{-1} \text{ K}^{-1}$), suffered from poor energy level mismatch alignment with perovskites, difficulty in work function tuning, and poor interfacial contact with perovskites, which readily formed recombination centers^{23,24}. Similarly, CuO_x exhibited high thermal conductivity ($2\sim 3 \text{ W m}^{-1} \text{ K}^{-1}$), but its narrow bandgap ($< 2 \text{ eV}$) and low transmittance in the long-wavelength region greatly limited the short-circuit current densities (J_{sc}) of PSCs²⁵. Therefore, spinel-type oxides represented a promising class of HTMs that simultaneously offer high thermal conductivity and suitable optoelectronic properties. However, since the energy levels of previously reported spinel-type HTMs did not match with perovskite films, it was necessary to develop special spinel-type HTMs coupling with suitable energy level.

To identify the most suitable spinel-type oxide, we then synthesized various materials (CuFe_2O_4 , CuBi_2O_4 , CuCo_2O_4) through hydrothermal synthesis method, respectively. For CuFe_2O_4 , it was found that Fe^{3+} was prone to valence variation (Supplementary Fig. 3), leading to relatively poor lattice stability. Meanwhile, CuFe_2O_4 had a low valence band maximum (E_{VB}), leading to a large hole extraction barrier²⁶. For CuCo_2O_4 , it had a poor solution dispersibility due to the agglomeration of nanocrystals (Supplementary Fig. 4), which was detrimental to the uniformity of the substrate film. Additionally, Co^{3+} exhibited strong oxidizability and could readily incorporate into the perovskite lattice, inducing the formation of defects²⁷⁻²⁹. Other spinel-type oxides (such as

NiFe₂O₄, ZnFe₂O₄) suffered from low hole mobility and serious energy level mismatch³⁰. In contrast, CuBi₂O₄ achieved a superior balance among heat conduction, hole transport and stability, which was regarded as an ideal HTM for managing heat conduction in PSCs³¹.

The crystallinity of CuBi₂O₄ nanocrystals (NCs) was examined by X-ray diffraction (XRD) measurement, where the highest intensity of diffraction peaks was indexed to the (211) plane of CuBi₂O₄ NCs according to the PDF#48-1886, and there was no additional diffraction peaks derived from impurities (Fig. 1b). Transmission electron microscopy (TEM) images further confirmed its lattice structure and morphology. CuBi₂O₄ NCs was cylindrical with an average particle size of 52 nm and an average thickness of 35 nm (Fig. 1c and Supplementary Fig. 5). The morphology and lattice plane of CuBi₂O₄ NCs could also be observed by high-resolution TEM images. The *d*-spacing of 3.2 Å was determined to be the (211) lattice plane (Fig. 1d), indicating that the as-synthesized CuBi₂O₄ NCs have high crystallinity, matched well with the XRD result. We further evaluated the batch-to-batch reproducibility of the NCs' size and morphology. Supplementary Fig. 6 showed that the final products all exhibited uniform nanocrystal morphology with no obvious differences between batches. In addition, the corresponding fast Fourier transform (FFT) pattern on a single NC confirmed its single crystalline nature (Supplementary Fig. 7)³². To investigate the elemental composition, we conducted X-ray photoelectron spectroscopy (XPS). The core-level spectra of Cu 2_p_{3/2}, Bi 4_f_{7/2}, and O 1_s were located at 931.9, 158.9 and 529.0 eV (Supplementary Fig. 8), corresponding to the Cu²⁺, Bi³⁺ and O²⁻, respectively. Extra O 1_s peaks at 530.9 and 532.0 eV originated from the oxygen vacancy and surface adsorbed oxygen (-OH)³³. Elemental mapping images further confirmed the elemental composition of the single CuBi₂O₄

NC (Supplementary Fig. 9), indicative of the high purity of CuBi₂O₄ NCs.

Moreover, we found that the onset temperatures at 3% weight loss for CuBi₂O₄ NCs were over 800 °C (Supplementary Fig. 10) by thermogravimetry (TG) measurement, indicative of the extremely outstanding thermal stability, far exceeding that of the previously reported inorganic HTMs. Ultraviolet photoelectron spectroscopy (UPS) and ultraviolet-visible (UV-vis) absorption spectroscopy were subsequently measured to study the band position of CuBi₂O₄³⁴. Supplementary Fig. 11 showed that the energy of band gap (E_g), E_{VB} , conduction band minimum (E_{CB}), and the Fermi level (E_F) for the CuBi₂O₄ substrate film were calculated to be 2.09, -5.11, -3.02 and -4.14 eV, respectively. The corresponding energy difference between E_F and E_{VB} was 0.97 eV, indicative of a p-type feature for CuBi₂O₄. These E_{VB} and E_g values were suitable to serve as a HTM in inverted PSCs. Space charge-limited current (SCLC) measurement (Supplementary Fig. 12 and Supplementary Note 1) also verified that CuBi₂O₄ NCs had a better hole mobility (μ_{hole}) of $3.65 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. Overall, we successfully synthesized CuBi₂O₄ NCs with suitable energy level, high crystallinity, excellent thermal stability and remarkable optoelectronic properties.

We further examined the thermal conductivity of CuBi₂O₄ by TCA measurement. Fig. 1e showed that CuBi₂O₄ exhibited a suitable thermal conductivity of $1.522 \text{ W m}^{-1} \text{ K}^{-1}$ at 300 K, and the thermal conductivity did not change significantly even if the temperature rose to 400 K. Moreover, the thermal diffusion coefficient (α) of CuBi₂O₄ was determined to be $0.828 \text{ mm}^2 \text{ s}^{-1}$ at 300 K, whereas the heat capacity (C_p) of CuBi₂O₄ could be calculated to be $0.185 \text{ J g}^{-1} \text{ K}^{-1}$ (Supplementary Table 3 and Supplementary Note 2). These results indicated that CuBi₂O₄ had excellent heat-transfer properties. Density functional theory (DFT) calculations were further

conducted to confirm the corresponding thermal conductivity. Fig. 1f showed the electronic thermal conductivity (κ_e) and the lattice thermal conductivity (κ_l) of CuBi_2O_4 as a function of temperature. Due to the low value of Lorentz number and electrical conductivity, the κ_e value was negligible and the κ_l value was crucial to the thermal conductivity of CuBi_2O_4 . At room temperature of 300 K, the value of κ_e was only $0.1926 \text{ W m}^{-1} \text{ K}^{-1}$ and the value of κ_l was $1.33 \text{ W m}^{-1} \text{ K}^{-1}$. The thermal conductivity of CuBi_2O_4 could be calculated to be $1.5226 \text{ W m}^{-1} \text{ K}^{-1}$ (Supplementary Table 4 and Supplementary Note 3). Hence, it was promising to construct the DCZ-4P@ CuBi_2O_4 composite HTL to tune heat-transfer properties.

Featuring double phosphoric acid groups in DCZ-4P, it could combine with CuBi_2O_4 through coordination bonds and anchor on the fluorine-doped tin oxide (FTO) substrate (Fig. 1g). Fourier transform infrared (FT-IR) spectroscopy confirmed this interaction. The characteristic P=O stretching vibration of DCZ-4P located at 1226.1 cm^{-1} was shifted to 1235.4 cm^{-1} for the DCZ-4P@ CuBi_2O_4 sample (Fig. 1h), indicating the successful combination behavior between DCZ-4P and CuBi_2O_4 . Besides, atomic force microscopy-infrared spectroscopy (AFM-IR) was further employed, where the DCZ-4P@ CuBi_2O_4 substrate film showed a uniform infrared signal intensity at 1235 cm^{-1} , corresponding to the P=O group (Fig. 1i), which indicated that DCZ-4P@ CuBi_2O_4 could anchor on the FTO substrate. This interaction was expected to increase the interfacial binding energy and reduce the reflection and scattering of phonons at the buried interface.

Phonon-engineered heat conduction.

To tune the thermal conductivity, we subsequently optimized the molar ratio of DCZ-4P and CuBi_2O_4 . It was found that the thermal conductivity of the DCZ-4P@ CuBi_2O_4 composite HTL

could be optimized to 0.875, 1.154 and 1.322 W m⁻¹ K⁻¹ for 2:1, 1:1 and 1:2 DCZ-4P@CuBi₂O₄ (Supplementary Fig. 13) at 300 K, respectively. The 1:1 DCZ-4P@CuBi₂O₄ substrate film also exhibited the suitable energy level (Supplementary Fig. 14 and 15) and photoelectric properties (Supplementary Fig. 16), which could be the best choice to use in PSCs. Supplementary Fig. 17 demonstrated that the DCZ-4P@CuBi₂O₄-based substrate film exhibited the highest conductivity when compared to the DCZ-4P and DCZ-4P@NiO_x substrate film, even at 85 °C. Based on the 1:1 DCZ-4P@CuBi₂O₄ substrate film, a better thermal compatibility between the CuBi₂O₄-based composite HTL and perovskite films was achieved (Fig. 2a and Supplementary Table 5).

DFT calculations were conducted again to provide qualitative insight into the possible vibrational origin of the increased thermal conductivity. Given that the poor intramolecular electron delocalization and a limited number of free electrons, DCZ-4P was unable to significantly transfer heat. Instead, heat transfer primarily occurred between molecules through molecular vibrations (e.g., phonons). Thus, the thermal conductivity of DCZ-4P relied on the contribution of lattice thermal conductivity (Supplementary Fig. 18). We further examined the phonon properties of DCZ-4P, DCZ-4P@NiO_x and DCZ-4P@CuBi₂O₄. Based on the idealized model, an in-plane-periodic slab model (2-3 atomic layers thick) with vacuum layer (≥ 15 Å) in the z-direction was adopted for the calculations (Supplementary Fig. 19), in which DCZ-4P contained a phosphorus-based anchoring group and a flexible alkyl chain. Due to the weak van der Waals forces existing between these isolated carbazole molecules, there was a weak dispersion on the low-frequency phonons, and it was basically flat at high frequencies (Fig. 2b). When a heterojunction interface was formed, the dispersion on the low-frequency phonons underwent a very strong correction,

indicating that there was a very strong anchoring effect at DCZ-4P and the interface (Fig. 2c-d). These anchoring effects had changed the group velocity and lifetime of the phonons, thereby affecting thermal conductivity. Supplementary Fig. 20 showed that the lattice thermal conductivity did not undergo significant changes when the k -point mesh was changed ($3\times3\times1$ and $5\times5\times1$), suggesting the calculated thermal conductivity was converged well. In the idealized model, DCZ-4P@CuBi₂O₄ showed a higher lattice thermal conductivity than DCZ-4P and DCZ-4P@NiO_x over the temperature range of 50-800 K. Additionally, the maximum group velocity ($v_{\max}$) of DCZ-4P@CuBi₂O₄ reached 341.23 m s⁻¹ (Fig. 2e-g), whereas the group velocity of DCZ-4P and DCZ-4P@NiO_x in the 0-20 THz range were only 59.98 and 147.36 m s⁻¹, respectively. At the same time, the critical lifetime of phonons at different frequencies showed a similar trend, particularly evidenced that the change in thermal conductivity was mainly attributed to low-frequency phonons, which could be regarded as the acoustic phonon³⁵. This low-frequency acoustic phonon in spinel-type oxides were known to have longer lifetimes and higher group velocities, thus contributing more to the overall lattice thermal conductivity than optical phonons³⁶. These idealized calculations suggested that the CuBi₂O₄-containing model might favor low-frequency acoustic phonon propagation.

Unfortunately, the as-constructed DCZ-4P@CuBi₂O₄ composite model was intrinsically non-crystalline and interfacial in nature. It possessed multiple interfaces (e.g., DCZ-4P/CuBi₂O₄, film/substrate, particle/matrix), which constituted one of bottlenecks for heat transfer: (1) The composite film contained a large number of interfaces, forming an "interface network" through which phonons must undergo multiple scattering events to complete heat transfer³⁷. (2) Thin

amorphous interfacial layers (>2 nm) could increase phonon scattering, thereby reducing thermal conductivity³⁸. (3) Interface defects could give rise to distinctive high-frequency vibrational modes that, under certain conditions, enhanced interfacial heat conduction³⁹. These factors would affect the actual calculation results, and they could not accurately reflect the actual thermal conductivity changes of the DCZ-4P@CuBi₂O₄ composite layer. Nevertheless, the idealized calculations might still provide qualitative insight into how interfacial coordination affected low-frequency vibrational dispersion and local vibrational coupling. It was worth noting that the isolated DCZ-4P model was used for qualitative explanation of intrinsic low thermal conductivity rather than quantitative comparison with composite models. Additionally, the original intention of our calculation was to qualitatively reveal the regulation trend of interfacial interaction on phonon transport behavior, rather than to quantitatively predict the absolute value of thermal conductivity. In the future, constructing appropriate disordered/amorphous models which captured the essential structural features and employing advanced theoretical approaches (e.g., molecular dynamics simulation) could account for the effects of amorphousness and interfaces on actual heat transfer.

To better rationalize the enhanced phonon group velocities, we referred to the theoretical correlation between phonon group velocities and elastic moduli of the material. The measured Young's modulus of the DCZ-4P, DCZ-4P@NiO_x and DCZ-4P@CuBi₂O₄ samples was 4.48, 43.13 and 154.72 GPa (Supplementary Fig. 21), respectively. Then, the shear modulus could be calculated to be 1.68, 16.22 and 59.97 GPa for the DCZ-4P, DCZ-4P@NiO_x and DCZ-4P@CuBi₂O₄ samples, respectively. According to the longitudinal and transverse wave velocity equation (Supplementary Note 4), yielding the corresponding longitudinal group velocity (v_L) and

transverse group velocity (v_T)⁴⁰. The resulting average group velocity (v_{ave}) of the DCZ-4P, DCZ-4P@NiO_x and DCZ-4P@CuBi₂O₄ samples was 1682.53, 3007.93 and 3821.81 m s⁻¹ (Supplementary Table 6), respectively. The DCZ-4P@CuBi₂O₄ sample showed the highest group velocity, consistent with the trend from DFT calculations. Notably, the DFT-calculated phonon group velocities referred to low-frequency acoustic phonons in the periodic slab model, while the experimental values derived from elastic moduli represent macroscopic phonon acoustic velocities. Only the relative trends, not absolute values, were compared. Overall, the improved thermal conductivity and reduced overall thermal resistance should be attributed to multiple coupled factors, including CuBi₂O₄ incorporation, improved interfacial adhesion, regulated film morphology, thermal-expansion compatibility, and possible low-frequency acoustic phonon propagation.

We further evaluated the thermal-expansion coefficient (α_e) of different films by thermomechanical analysis measurements. The mismatched thermal-expansion coefficient between perovskites and HTLs would cause heat and/or strain accumulation. By comparing these values, we found that there was a significant difference between perovskites and traditional HTLs (Fig. 2h), which negatively affected the shrinkage of perovskites during annealing process. Notably, there existed a well-matched alignment distribution of thermal-expansion coefficients between perovskites and the DCZ-4P@CuBi₂O₄ HTL. This alignment distribution derived from the regulated heat conduction would release residual strain and induce the formation of high-quality perovskite films⁴¹. The overall thermal resistance (R_{th}) of different devices (the only DCZ-4P sample was defined to be control, the 1:1 DCZ-4P@NiO_x and 1:1 DCZ-4P@CuBi₂O₄ samples

were abbreviated as NiO_x and CuBi_2O_4 , respectively) could be further obtained by thermal interface material (TIM) tester, where the CuBi_2O_4 -based device showed the lowest value of $1.298 \times 10^{-2} \text{ m}^2 \text{ K W}^{-1}$ (Supplementary Fig. 22) compared to the control ($1.926 \times 10^{-2} \text{ m}^2 \text{ K W}^{-1}$) and NiO_x -based ($1.812 \times 10^{-2} \text{ m}^2 \text{ K W}^{-1}$) devices. It was attributed to enhanced heat conduction and optimal thermal-expansion coefficient alignment. Interestingly, the device based on the solely of CuBi_2O_4 layer exhibited a higher R_{th} of $1.625 \times 10^{-2} \text{ m}^2 \text{ K W}^{-1}$ than that of the CuBi_2O_4 -based device. This might be related to unfavorable interfacial contact, increased interfacial scattering, and possible defect traps in the solely of CuBi_2O_4 layer¹⁸. These results demonstrated that the CuBi_2O_4 -based sample exhibited a more exceptional thermal compatibility than that of the control and NiO_x -based samples.

To study the heat transfer dynamics on different substrate films, we first carried out COMSOL simulations based on finite element method to investigate heat transfer dynamics. As shown in Fig. 2i, the heat transfer on the CuBi_2O_4 -based substrate showed a faster temperature enhancement from the center (heat source) to the outside (cold source) compared to the control and NiO_x -based substrate films (Supplementary Fig. 23)¹². It could be seen that the heat source ($> 50^\circ\text{C}$) of the CuBi_2O_4 -based composite HTL had spread to 3 mm at 1.7 s (Supplementary Fig. 24), while the control and NiO_x -based films had only spread to 1 mm under the same conditions. It demonstrated that the CuBi_2O_4 -based substrate film achieves a more efficient heat transfer process, which was conducive to rapidly heating the substrate and accelerating the nucleation rate of perovskites. Based on the optimized heat conduction, we further used infrared (IR) camera to examine IR signal of the as-deposited perovskites. It was found that the CuBi_2O_4 -based perovskite film has a faster

heating rate and could be heated to the specified temperature of 100 °C in just 350 s, while the control and NiO_x-based perovskite film required over 450 s (Fig. 2j, Supplementary Fig. 25 and 26). This phenomenon suggested that the CuBi₂O₄-based perovskite film was more conducive to annealing perovskites and promoting its nucleation, crystallization and growth process.

Perovskite crystallization dynamics and surface morphologies.

As mentioned, heat-transfer and thermal expansion dynamics would significantly affect the perovskite crystallization dynamics. In-situ UV-vis absorption spectroscopy was then employed to study the nucleation and crystallization dynamics of perovskite films^{1,42}. The CuBi₂O₄-based perovskite film exhibited a steeper absorption intensity after dropping anti-solvent at 35 s compared to the control and NiO_x-based perovskite films (Fig. 3a-c), respectively, indicative of a relative fast nucleation rate for the CuBi₂O₄-based perovskite film. As confirmed by the extracted UV-vis absorption at 420 nm (Supplementary Fig. 27), a faster climbing rate of UV-vis absorption intensity was observed for the CuBi₂O₄-based perovskite film compared to the control and NiO_x-based perovskite films. We inferred that the introduction of CuBi₂O₄ NCs promoted heterogeneous-nucleation behavior, resulting in accelerating the formation of nucleation sites (Supplementary Fig. 28)⁴³. For the thermal annealing process, although all films reached the highest peak intensity almost at the same time, a steeper absorption climbing rate could be clearly observed in the CuBi₂O₄-based perovskite film during the initial stage. This was attributed to the fact that enhanced heat conduction in the CuBi₂O₄-based perovskite film could accelerate the initial nucleation rate.

In-situ photoluminescence (PL) spectroscopy showed similar trends in perovskite nucleation

and crystallization process (Fig. 3d-f). For the spin-coating process, the CuBi_2O_4 -based perovskite film exhibited a stronger PL intensity after stripping the anti-solvent, which could be evidenced by the increased intensity at UV-vis spectra. For the annealing process, it was divided into three crystallization stages⁴⁴. In the stage I, the nucleation and crystal growth of perovskites coexisted after heating the substrates, which greatly affected by heat conduction. The PL signal showed that the CuBi_2O_4 -based perovskite film rapidly grew to its maximum value. This might be the fact that the enhanced heat conduction promoted amorphous perovskite films rapidly absorbed heat, resulting in the formation of a large number of nucleation sites. In stage II, the PL signal began to decline gradually due to the crystal restructuring⁴⁵. The longer time suggested slower and more thorough crystal reconstruction behavior, where the CuBi_2O_4 -based perovskite film exhibited a longer time of 23.4 s than those of the control (7.4 s) and NiO_x -based (13.6 s) perovskite films (Supplementary Fig. 29). In stage III, PL intensity of these perovskite films all showed a tendency to increase slowly derived from the recrystallization process. The CuBi_2O_4 -based perovskite film exhibited the slowest crystallization process. This was attributed to the interfacial chemistry and ionic interactions between CuBi_2O_4 -based composite and perovskite layers. To be specific, there was a possible Bi-related interfacial interaction due to Bi^{3+} with a similar ionic radius to Pb^{2+} (Supplementary Fig. 30)⁴⁶. This factor could reduce the crystallization driving force, retard re-born nucleus growth and grain coalescence, giving rise to a delayed increase in PL intensity and a slow perovskite phase transition. Additionally, the control perovskite film exhibited a trace amount of decomposition signal, indicative of the poor stability. These results demonstrated that the CuBi_2O_4 -based perovskite film exhibited a faster nucleation rate and a slower crystallization process than

that of the control and NiO_x-based perovskite films, which was beneficial for achieving high-quality of perovskite films with low defect density.

We measured the surface hydrophilicity of different substrate films. It was found that the CuBi₂O₄-based substrate film exhibited a better wettability to the perovskite precursor solution (Supplementary Fig. 31), indicating that the lower surface energy in the CuBi₂O₄-based substrate film. To study the effect of enhanced heat conduction on the quality of perovskite films, we conducted scanning electron microscopy (SEM) and atomic force microscopy (AFM)³⁴. As shown in Fig. 3g and Supplementary Fig. 32, the CuBi₂O₄-based perovskite film presented a larger grain size (Supplementary Fig. 33), a smoother surface and lower roughness (Supplementary Fig. 34) than that of the control and NiO_x-based perovskite film whether at the buried interface or on the top surface. The corresponding cross-section SEM images further demonstrated that there were almost no interstitial traps in the CuBi₂O₄-based perovskite film (Supplementary Fig. 35). We also observed that the residual PbI₂ was enriched at the grain boundaries of the control top-perovskite film, attributed to the decomposition of perovskites caused by heat accumulation⁴⁷. As confirmed by XRD patterns, there was a diffraction peak at 13.2° corresponding to PbI₂ in the control and NiO_x-based top-perovskite film (Supplementary Fig. 36), whereas the CuBi₂O₄-based top-perovskite film had no signals, indicative of the better stability^{48,49}. Additionally, the CuBi₂O₄-based perovskite film showed an enhanced diffraction intensity at 14° and 28° corresponding to (100) and (200) plane of α -phase perovskite (Supplementary Fig. 37) compared to the control and NiO_x-based perovskite films. More importantly, we found that the (100) plane in FA-based perovskites closely matched the (211) plane in CuBi₂O₄ (Supplementary Fig. 38), which

contributed to forming the out-of-plane orientation of perovskites along (100) plane.

In order to study the residual strain, grazing-incidence X-ray diffraction (GIXRD) was measured⁵⁰. The depth-dependent crystal structure information after thermal annealing was shown in Supplementary Fig. 39. The diffraction peaks of the control perovskite film at $2\theta = 28.2^\circ$ exhibited a significant shift toward lower angles, whereas the NiO_x-based and CuBi₂O₄-based perovskite films exhibited slight shifts. These specific offsets could be converted into the value of residual strain (σ_M) by fitting the line of $2\theta - \sin^2\psi$ ⁵¹. The corresponding σ_M was calculated to be 26.51, 13.63 and 12.08 MPa according to the slope of -0.0496, -0.0255 and -0.0226 for the control, NiO_x-based and CuBi₂O₄-based perovskite films (Fig. 3h and Supplementary Note 5), respectively, indicative of the released residual strain in the CuBi₂O₄-based perovskite film, attributed to the optimal thermal-expansion coefficient alignment. These results demonstrated that the high-quality of perovskite films could be achieved by introducing CuBi₂O₄ to enhance heat conduction, which was conducive to improving carrier transport and suppressing trap-assisted nonradiative recombination. Depth-resolved grazing-incidence wide-angle X-ray scattering (GIWAXS) was further employed to evaluate the crystallographic structure of different perovskite films⁵². Compared to the control and NiO_x-based perovskite films, brighter diffraction rings at q_{xy} values of 1 and 2 Å⁻¹ corresponding to the (100) and (200) crystal planes could be observed in the CuBi₂O₄-based perovskite film (Fig. 3i). Additionally, the CuBi₂O₄-based perovskite film showed a strongly enhanced signal at the polar angle (χ) = 38° or 68° of (100) plane (Fig. 3j), indicative of an oriented crystallization. These results demonstrated that the CuBi₂O₄-based perovskite film featured high crystallinity, large grain size and ordered orientation.

Carrier transport and recombination dynamics.

Given that both light absorption and heat excitation predominantly originated from the bottom side of PSCs, the poor thermal compatibility at the buried interface between adjacent layers could give rise to rapid local heat accumulation. This fostered the formation of local hotspots and accelerated the injection of hot carriers (charge-carriers excited above the bandgap). The rapid injection behavior would increase the recombination probability of hot carriers and the breakdown risk of devices, and could generate excessive heat¹⁴. On account of the fact that hot carriers usually increased the energy losses in the form of heat to naturally waste the excess photon energy before cooling down to lattice temperature¹⁵. Theoretically, delaying the relaxation of hot carriers by regulating heat conduction was expected to make full use of photon energy and suppress nonradiative recombination.

Enhanced heat conduction would avoid the rapid injection of hot carriers, thereby improving the utilization of high-energy photogenerated carriers. We performed the femtosecond transient absorption spectroscopy (fs-TAS) to gain insight into the relaxation of hot carriers in detail, as shown in Fig. 4a-c. By using high-energy pump light (400 nm) excitation, we observed a broadened bleaching signal at 1.65-1.75 eV in the normalized TA spectra (Fig. 4d-f). This energy region was called the energy tail, representing the hot-carrier relaxing process, which was caused by the initial non-equilibrium carrier distribution that quickly reached the Fermi-Dirac distribution⁵³. According to fitting the high-energy tail, it disappeared more slowly in the CuBi₂O₄-based perovskite film than that of the control and NiO_x-based perovskite films (Fig. 4g). The corresponding relaxation time of hot carriers was significantly delayed to 759 fs for the CuBi₂O₄-

based perovskite films compared to the control (353 fs) and NiO_x-based (186 fs) perovskite films. The prolonged lifetime of hot carriers manifested that the CuBi₂O₄-based perovskite film slowly lost their excess energy of hot carriers before they thermalized to the perovskite crystal lattice compared to the control perovskite film, which was propitious to the extraction of hot carriers and had a positive effect on the device performance. Moreover, the energy loss rate of carriers (J_r) was slowly decreased from 0.3429 eV ps⁻¹ for the control perovskite film to 0.1219 and 0.0351 eV ps⁻¹ for the NiO_x-based and CuBi₂O₄-based perovskite films, respectively, when the temperature of hot carriers (T_c) was less than 2000 K (Fig. 4h and Supplementary Note 6). This result demonstrated that excess photon energy carried by hot carriers in the control and NiO_x-based perovskite film was unintentionally wasted, while the utilization factor had been increased by approximately 10-fold for the CuBi₂O₄-based perovskite film.

In addition to the hot carrier, we also studied the ordinary carrier transport dynamics. When using low-energy (600 nm) pump-light, we obtained the lifetime of ordinary carriers⁵⁴. Supplementary Fig. 40 showed 2D pseudo-color TA plots of different perovskite films, where the lifetime of the CuBi₂O₄-based perovskite film was calculated to be 3765.7 ps, while that of the control and NiO_x-based perovskite films was only 2008.0 and 2132.2 ps (Supplementary Note 7 and Supplementary Table 7), respectively. It meant that the lifetime of perovskite film was prolonged and the defect density was significantly reduced in the CuBi₂O₄-based perovskite film. Steady-state photoluminescence (PL) and the time-resolved photoluminescence (TRPL) measurements further verified these findings (Supplementary Fig. 41, Supplementary Note 8 and Supplementary Table 8)⁵⁵. To detect the carrier collection times (τ_c) and recombination times (τ_{rec}),

we measured intensity-modulated photocurrent spectroscopy (IMPS) and intensity-modulated photovoltage spectroscopy (IMVS)⁵⁶. The value of τ_c was calculated to be 0.86, 0.80 and 0.73 μs at 10% irradiation intensity (Supplementary Fig. 42 and Supplementary Note 9) for the control, NiO_x -based and CuBi_2O_4 -based devices, respectively. It indicated that the CuBi_2O_4 -based device had an enhanced carrier collection behavior even under weak light, which was beneficial for manufacturing indoor photovoltaic devices. Meanwhile, the value of τ_{rec} for the CuBi_2O_4 -based device was calculated to be 5.49 μs at 10% irradiation intensity (Supplementary Fig. 43 and Supplementary Table 9), higher than that of the control (3.96 μs) and NiO_x -based (5.37 μs) devices, indicative of the inhibited nonradiative recombination in the CuBi_2O_4 -based device. More importantly, The CuBi_2O_4 -based device demonstrated the shortest τ_c and the longest τ_{rec} whatever at different irradiation intensities (10%~80% light-emitting diode). The related charge collection efficiency (η_{cc}) could be extracted from the τ_c and τ_{rec} values, where the CuBi_2O_4 -based device all exhibited larger values than those of the control and NiO_x -based devices (Supplementary Fig. 44), especially demonstrating a maximal η_{cc} of 86.74% at 10% irradiation intensity.

We further measured electrochemical impedance spectroscopy (EIS) to evaluate the changes in resistance at the operating temperature (Supplementary Fig. 45). As shown in Fig. 4i and Supplementary Table 10, the series resistance (R_s) of these devices increased gradually with rising temperature, whereas the recombination resistance (R_{rec}) decreased accordingly. Furthermore, the CuBi_2O_4 -based device always exhibited the lowest R_s and the largest R_{rec} when the temperature ranged from 20 °C to 80 °C. These results demonstrated that faster carrier collection and inhibited nonradiative recombination in the CuBi_2O_4 -based device. This was attributed to the high-quality

of perovskite films and the delayed relaxation process of hot carriers. Transient photocurrent (TPC) and transient photovoltage (TPV) measurements were then performed (Supplementary Fig. 46). The CuBi_2O_4 -based device showed a fast photocurrent decay time (τ_{TPC}) of 2.6 μs and a prolonged photovoltage decay time (τ_{TPV}) of 93.2 μs (Supplementary Table 11 and 12), which were better than that of the control and NiO_x -based devices. These results indicated that accelerated charge transfer and inhibited nonradiative recombination, consistent with the results of IMPS and IMVS⁵⁷. We then compared the hole mobility and electronic mobility of different devices. Supplementary Fig. 47 showed the hole mobility was significantly improved in the CuBi_2O_4 -based device, resulting in a balance of electron and hole transport.

Additionally, the V_{OC} and J_{SC} versus logarithm of the light density curves were measured to extract the ideality factor of different devices (Supplementary Fig. 48 and Supplementary Note 10). The CuBi_2O_4 -based device had a smaller ideality factor of 1.33 than that of the control (1.45) and NiO_x -based (1.41) devices, indicative of the reduced trap-assisted nonradiative recombination. Mott-Schottky curves of PSCs were also measured, Supplementary Fig. 49 showed that the CuBi_2O_4 -based device obtained a larger value (1.15 V) of built-in potential (V_{bi}) than that of the control (1.05 V) and NiO_x -based (1.08 V) devices. The larger V_{bi} driven by the built-in electric field indicated that the efficient carrier separation and transport process⁵⁵. SCLC measurement was employed again to quantify the trap-state density (N_{trap})⁵⁸. As shown in Supplementary Fig. 50 and Supplementary Note 11, the N_{trap} of the CuBi_2O_4 -based device was calculated to be $4.95 \times 10^{15} \text{ cm}^{-3}$, which was lower than that of the control ($6.08 \times 10^{15} \text{ cm}^{-3}$) and NiO_x -based ($5.27 \times 10^{15} \text{ cm}^{-3}$) devices. It meant that the inhibited carrier recombination in the CuBi_2O_4 -based device,

contributed to achieving high V_{OC} and FF. The results of deep-level transient spectroscopy (DLTS) measurements further demonstrated that the CuBi_2O_4 -based device exhibited the lowest deep-level defect concentration ($7.32 \times 10^{14} \text{ cm}^{-3}$) when compared to the control ($7.60 \times 10^{14} \text{ cm}^{-3}$) and NiO_x -based ($8.48 \times 10^{14} \text{ cm}^{-3}$) devices (Supplementary Fig. 51 and Supplementary Note 12), which corroborated the SCLC results. External quantum efficiencies of electroluminescence (EQE-EL) were measured to check the V_{OC} loss derived from nonradiative recombination $(\Delta V_3)^{59}$. The values of EQE_{EL} were calculated to be 1.12%, 3.49% and 5.58% for the control, NiO_x -based and CuBi_2O_4 -based devices (Supplementary Fig. 52), respectively, corresponding to the ΔV_3 of 116, 86 and 74 mV according to the detailed balance theory (Supplementary Note 13). These results collectively demonstrated that the CuBi_2O_4 -based device exhibited the accelerated carrier transport and inhibited nonradiative recombination, which was beneficial for improving efficiency and operational stability of devices.

Photovoltaic performance of devices.

The inverted PSCs with an architecture of FTO/DCZ-4P@ CuBi_2O_4 /Cs_{0.05}MA_{0.05}FA_{0.90}PbI₃/C₆₀/BCP/Ag was fabricated to study the effect of enhanced heat conduction on the photovoltaic performance. As expected, the devices based on the 1:1 DCZ-4P@ CuBi_2O_4 substrate film showed the champion PCE (Supplementary Fig. 53 and Supplementary Table 13). Meanwhile, we determined that the CuBi_2O_4 -based device with a concentration of approximately 0.5 mg mL^{-1} could obtain the highest PCE (Supplementary Fig. 54 and Supplementary Table 14). The champion CuBi_2O_4 -based device (Fig. 5a and Supplementary Table 15) delivered a high PCE of 27.18% ($V_{OC} = 1.188 \text{ V}$, $J_{SC} = 26.04 \text{ mA cm}^{-2}$ and FF = 87.87%),

whereas the PCE of the control device was 25.91% ($V_{OC} = 1.173$ V, $J_{SC} = 25.92$ mA cm⁻² and FF = 85.20%). This best PCE was certified to be 26.83% (Supplementary Fig. 55) from National Photovoltaic Product Quality Inspection & Testing Center, China, which was one of the highest efficiencies currently reported for inverted PSCs (Supplementary Table 16). Meanwhile, the stable power output efficiency (SPO) for the control, NiO_x-based and CuBi₂O₄-based devices was estimated to be 25.88% at 1.06 V (Supplementary Fig. 56), 26.05% at 1.06 V and 26.62% at 1.08 V (certified 26.51%), respectively. The hysteresis effect index (HEI) of the champion CuBi₂O₄-based device was calculated to be only 2.94% (Supplementary Fig. 57 and Supplementary Note 14), which was lower than that of the control (5.40%) and NiO_x-based (3.93%) device. To calculate the detailed integrated photocurrent densities of PSCs, external quantum efficiency (EQE) spectrum was then measured. The control, NiO_x-based and CuBi₂O₄-based devices showed an integrated photocurrent density of 25.24, 25.30 and 25.55 mA cm⁻² (Supplementary Fig. 58), respectively, which matched well with the values extracted from the J - V curves. Moreover, the excellent repeatability of CuBi₂O₄-based devices was verified by fabricating 40 individual PSCs (Fig. 5b), which yielded an average PCE of 26.98±0.20%, higher than those of the control (25.89±0.17%) and NiO_x-based (26.20±0.24%) devices.

In addition, the device consisting solely of the CuBi₂O₄ layer achieved an optimal PCE of 24.13% ($V_{OC} = 1.161$ V, $J_{SC} = 25.48$ mA cm⁻², and FF = 81.55%). Unfortunately, the poor coverage of CuBi₂O₄ NCs led to a direct contact between the perovskite and FTO substrates, which limited the improvement of efficiency. However, it remained one of the highest records for PSCs based on spinel-type layers to date (Supplementary Table 17). To assess the validity of enhanced heat

conduction, we fabricated large-area and indoor PSCs (Fig. 5c, Supplementary Fig. 59 and 60, Supplementary Table 18). The CuBi_2O_4 -based device delivered higher PCEs of 25.75% (1 cm^2), 19.01% (210 cm^2) and 44.07% (indoor, 1000 lux) than that of the control and NiO_x -based devices (Supplementary Fig. 61). Additionally, we constructed different devices based on different SAMs to confirm the uniqueness of DCZ-4P, where Me-4PACz-based and MeO-2PACz-based devices only obtained a lower PCE of 25.03% and 24.24% (Supplementary Fig. 62 and Supplementary Table 19) compared to the DCZ-4P-based device, respectively. These results demonstrated that highly efficient inverted PSCs could be achieved by constructing the DCZ-4P@ CuBi_2O_4 composite layer that aimed at balancing heat conduction.

To study the effect of enhanced heat conduction on the operational stability of PSCs, we directly examined IR signal of the whole cells using IR thermography. Due to the limited and scattered heat conducted by the light source, the heating rate of the CuBi_2O_4 -based device within 60 s was actually slower than that of the control and NiO_x -based devices (Supplementary Fig. 63). Moreover, the temperature of the whole CuBi_2O_4 -based device remained $41.5\text{ }^\circ\text{C}$ after 400 s (Supplementary Fig. 64). In contrast, the temperature of the control device had risen to $43.8\text{ }^\circ\text{C}$. The lower operating temperature of the CuBi_2O_4 -based device was conducive to improving the operational stability under practical application conditions and reducing the long-term operating cost. To assess the detailed heat dissipation behavior, we cooled down the device after continuous light soaking. Supplementary Fig. 65 and 66 showed that the CuBi_2O_4 -based device had a faster self-cooling rate than that of the control device, but the cooling process to room temperature took about the same time. We also let the device cool naturally from a high temperature of $85\text{ }^\circ\text{C}$ (Supplementary Fig.

67 and 68), the CuBi_2O_4 -based device cooled directly to room temperature in just 200 s, while the control and NiO_x -based sample took nearly 300 and 350 s, respectively. The short cooling time meant that the CuBi_2O_4 -based device could greatly extend the operational lifetime of PSCs in practical applications.

We then tracked the efficiency under continuous thermal aging at 85 °C in N_2 -filled glove box according to ISOS-D-2I. The CuBi_2O_4 -based device without encapsulation retained 90.1% of its initial efficiency after 2000 h storage, higher than that of the control (48.8%) and NiO_x -based (58.9%) devices (Fig. 5d). XRD patterns of perovskite films after aging 30 days at 85 °C confirmed the excellent thermal stability for the CuBi_2O_4 -based device, which was due to the high-quality perovskite film and enhanced heat conduction (Supplementary Fig. 69). Similarly, SEM images further demonstrated this advantage. Even after continuous annealing at 85 °C for 100 h, the CuBi_2O_4 -based perovskite film retained the black phase without decomposition (Supplementary Fig. 70). Damp-heat stability test (85 °C and 85% humidity in dark, ISOS-D-3) was also performed to evaluate the thermal stability of PSCs⁶⁰. The CuBi_2O_4 -based device with encapsulation remained over 90% of its initial PCE after aging 1000 h. Remarkably, it maintained 82.3% of its initial PCE after aging 2000 h (Fig. 5e), better than that of the control (47.6%) and NiO_x -based (53.5%) devices. To evaluate the thermal cyclic stability of PSCs, we conducted thermal cycling aging from room temperature to 85 °C following ISOS-T-1 every 6 h. The CuBi_2O_4 -based device without encapsulation maintained 85.6% of its initial PCE after 2000 h (Fig. 5f), while the control and NiO_x -based devices only remained 44.9% and 60% of their initial PCEs, respectively. For the light stability, we subjected them to continuous illumination (AM 1.5G, ISOS-L-1) at the

maximum power point (MPP). The CuBi_2O_4 -based device without encapsulation exhibited an excellent light stability, retaining 93.7% of its initial PCE after 2000 h of continuous MPP tracking (Fig. 5g). In contrast, the control and NiO_x -based devices only maintained 53.1% and 72.6% of their initial PCEs, respectively. Additionally, Supplementary Fig. 71 showed that the device based solely on the CuBi_2O_4 HTL without encapsulation maintained above 90% of its initial PCEs after light soaking for 500 h, indicative of the excellent operating stability of devices based on the only CuBi_2O_4 HTL. These results demonstrated that the CuBi_2O_4 -based device exhibited excellent operational stability.

Discussion

In summary, we report a previously undescribed phonon-engineered HTL, aiming to solve the problem of poor interlayer thermal compatibility, which differed from previous strategies that focused on the thermal conductivity of bulk perovskites or isolated heat-dissipating layers. We chelated DCZ-4P and CuBi_2O_4 that greatly enhances phonon group velocities, leading to an enhanced heat conduction and an optimal thermal-expansion coefficient alignment. Moreover, this advanced HTL substantially avoids heat accumulation, prolongs hot-carrier relaxation, tailors perovskite crystallization and suppresses nonradiative recombination. These synergistic effects culminate in exceptional device performance: a champion PCE of 27.18% (certified 26.83%), with high efficiencies maintained in scaled-up (25.75%, 1 cm^2 ; 19.01%, 210 cm^2) and indoor (44.07%, 1000 lux) configurations. Furthermore, the devices exhibit outstanding operational stability under a suite of stringent ISOS protocols (including ISOS-D-2I, ISOS-D-3, ISOS-T-1 and ISOS-L-1). We envision that the phonon-engineered strategy offers a generalizable pathway to reconcile the

trade-off between efficiency and stability, not only in perovskites but also in other emerging optoelectronic devices.

Methods

Materials

Unless otherwise noted, all chemicals were used as received without any further purification or modification. Copper(II) nitrate trihydrate ($\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$, 98%), bismuth(III) nitrate pentahydrate ($\text{Bi}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$, 98%), Iron(III) nitrate trihydrate ($\text{Fe}(\text{NO}_3)_3 \cdot 3\text{H}_2\text{O}$, 98%), Cobalt(II) nitrate hexahydrate ($\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, 98%) sodium hydroxide (NaOH, 98%), N,N-dimethylformamide (DMF, 99.9%), dimethyl sulfoxide (DMSO, 99.9%), anisole (99.7%), isopropyl alcohol (IPA, 99.5%), chlorobenzene (CB, 98%), ethanol (99.9%) and magnesium fluoride (MgF_2 , 99.99%) were purchased from Adamas-beta. Methylammonium iodide (MAI, 99.9%), lead(II) chloride (PbCl_2 , 99.9%) and methylamine hydrochloride (MAHCl , 99.9%) were purchased from Xi'an Yuri Solar Co., Ltd. Cesium iodide (CsI, 99.99%), formamidinium iodide (FAI, 99.5%), lead (II) iodide (PbI_2 , 99.999%), fullerene- C_{60} (99.9%) and bathocuproine (BCP) were obtained from Advanced Election Technology Co., Ltd. 9*H*,9'*H*-(3,3'-bicarbazole)-9,9'-diylbis(butane-4,1-diyl)diphosphonic acid (DCZ-4P, 99%) was purchased from TCI (Shanghai) Development Co., Ltd.

Synthesis of CuBi_2O_4 NCs

Firstly, 0.484 g $\text{Bi}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$, 0.12 g $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ and 0.174 g NaOH were added to 16 mL deionized water under stirring for at least 3 hours to obtain precursor solution. Then, the mixed solution was transferred in a stainless-steel autoclave with heating at 180 °C for 24 hours. The

precipitate was subsequently extracted by vacuum filtration and washed three times with anhydrous ethanol and deionized water, respectively, to remove unreacted substances and other impurities. The washed material was then dried at 60 °C for 24 hours to obtain a series of final products.

Perovskite precursor preparation

The $\text{Cs}_{0.05}\text{MA}_{0.05}\text{FA}_{0.90}\text{PbI}_3$ perovskite precursor was prepared by mixing PbI_2 (691.52 mg), FAI (232.16 mg), CsI (19.49 mg) and MAI (11.92 mg) in a solvent of 1 mL DMF and DMSO (4:1, v/v). Additional 5% PbI_2 and 15% MACl were added into the precursor solution. Stirring at room temperature at least 2 hours, and then filtered by a 0.22 μm polytetrafluoroethylene membrane before use.

Device fabrication

The patterned FTO glass substrates were cleaned via sonication in soap, deionized water, acetone and ethanol. 110 nm MgF_2 film was vacuum deposited on the back of FTO glass in nitrogen glovebox, then the cleaned FTO substrates were treated with ultraviolet ozone for 30 minutes before transferred into nitrogen glovebox. DCZ-4P and CuBi_2O_4 were dispersed in ethanol. DCZ-4P@ CuBi_2O_4 mixed ethanol precursor with different molar ratios (1:0, 2:1, 1:1, 1:2, 0:1) and concentrations (0~0.8 mg mL^{-1}), all these precursors were spin coated at 4000 rpm for 30 s (1000 rpm/s) and annealed at 100 °C for 30 minutes. After that, perovskite precursor solution was spread on the substrate and spin-coated at 1000 rpm for 10 s, followed by 5000 rpm for 30 s. During the final 7 seconds of the spin-coating process, 140 μL of anisole was dripped onto the substrate and annealed at 120 °C for 20 minutes. Then, propane-1,3-diammonium iodide (1 mg mL^{-1} in IPA and

CB mixed solution with 3:1 v/v) solution was dynamically spin-coated on perovskite at 5000 rpm for 25 s and annealed at 100 °C for 7 minutes. Finally, the devices were completed by thermal evaporation of C₆₀ (25 nm), BCP (6 nm) and Ag (110 nm). The active area of the devices was 0.09 cm².

Characterization

Current density-voltage (*J-V*) measurements were conducted using a Keithley 2400 source meter under AM 1.5G irradiation. Sunlight was generated using AAA class solar simulator with 300 W Xenon lamp, operated in a nitrogen glovebox. The *J-V* curves were extracted from 1.5 V to -0.1 V, with a voltage step of 0.02 V. A metal mask of 0.09 cm² was used for the *J-V* measurement. External quantum efficiency (EQE) measurements were performed by using Enlitech QE-R3011 EQE system. Maximum power point tracking (MPPT) measurement were conducted by using a commercial system supported by 91PVKSolar Co. Ltd, China. Other stability tests were performed under the ISOS-D-2I, ISOS-D-3 and ISOS-T-1 conditions, respectively. UPS and XPS measurements were carried out using an ESCALAB 250Xi system (Thermo Fisher Scientific) under ultrahigh vacuum conditions with a base pressure of 1×10^{-10} mbar. Transmission Electron Microscope (TEM) were tested on Talos F200S G2. X-ray diffraction (XRD) patterns were taken on Bruker AXS D8 advance with a Cu-K α X-ray irradiation source operated at 40 kV and 40 mA. Scanning electron microscopy (SEM) images were obtained by thermo scientific Apreo 2C. Atom force microscopy (AFM) images were observed on Bruker NanoIR3. The water contact angles (CA) were tested by SDC-350. UV-vis absorption spectra were tested on the UV-3600i Plus. Photoluminescence (PL) and Time-resolve PL (TRPL) spectra were measured by PicoQuant

FluoTime 510 nm laser excitation. Electrochemical impedance spectroscopy (EIS), Mott-Schottky (M-S), transient photovoltage spectrum (TPV) and transient current spectrum (TPC) were measured on Paios (Fluxim). Grazing incident wide-angle X-ray scattering (GIWAXS) measurements were performed by the XEUSS SAXS/WAXS equipment, at an incident angle of 0.5° . Grazing incidence X-ray diffraction (GIXRD) patterns were acquired in air by using a Rigaku Smartlab with Cu $K\alpha$ radiation ($\lambda = 1.5405 \text{ \AA}$). The femtosecond transient absorption measurements were performed on a Helios spectrometer (Ultrafast Systems) pumped by a Ti:sapphire regenerative amplifier (Legend Elite, Coherent) operating at 5 kHz with fundamental wavelength of 400 nm and pulse width of ~ 40 fs. The thermal resistance measurements were conducted by using a thermal interface material (TIM) tester (DRL-V) based on the steady-state one-dimensional heat flow method under controlled pressure and temperature conditions. To ensure experimental accuracy, the silver electrodes were removed from the complete devices prior to the measurements. All images were created by PowerPoint and VESTA.

Simulation methods

The structural relaxation was performed with the density functional theory (DFT) with the projector augmented wave (PAW) method implemented in the Vienna ab initio simulation package (VASP). We treat the exchange-correlation potential based on the Perdew–Burke–Ernzerhof generalized gradient approximation (PBE-GGA) functional. A $4 \times 4 \times 4$ k-point grid mesh density is used for geometry optimization. The cut-off energy is set at 450 eV. Energy convergence criteria of 1.0×10^{-7} eV/atom are set and Hellmann–Feynman forces are less than 0.01 meV/ $\text{\AA}$. Meanwhile, to better describe lattice thermal conductivity of DCZ-4P and DCZ-4P@CuBi₂O₄, the accurate

phonon frequencies are obtained by second-order force constants executed by VASP and Phonopy package and third-order force constant executed by VASP and Thirdorder software package. Finally, ShengBTE code was used to solve the phonon and electron Boltzmann transport equation (BTE). For the DCZ-4P model, DCZ-4P was simplified to carbazole for ease of calculation due to the molecular structure of DCZ-4P could not form a periodic unit cell. In contrast, a composite heterostructure was constructed for the DCZ-4P@NiO_x and DCZ-4P@CuBi₂O₄ samples (here, DCZ-4P contained a phosphorus-based anchoring group and a flexible alkyl chain), in which periodicity existed along the interface plane, allowing the phonon was well defined. The periodic interface enabled phonon scattering to have angle and frequency selectivity, and the wave vector (k) was conserved in the plane direction of the interface. To balance computational cost and structural stability, a two-layer oxide slab was used with the bottom layer fixed to mimic bulk constraints, while the surface layer and the DCZ-4P molecule were relaxed. Its adsorption configuration on the substrate was optimized through geometry relaxation and energy minimization.

Data availability

The data generated in this study are provided in the Supplementary Information/Source Data file. All other data are available from the corresponding authors on request. Source data are provided with this paper.

References

- [1] Xu, P. et al. Retarding the growth kinetics of chemical bath deposited nickel oxide films for efficient inverted perovskite solar cells and minimodules. *Adv. Mater.* **37**, e2505087 (2025).

- [2] Liu, S. et al. Buried interface molecular hybrid for inverted perovskite solar cells. *Nature* **632**, 536-542 (2024).
- [3] Du, J. et al. Face-on oriented self-assembled molecules with enhanced π - π stacking for highly efficient inverted perovskite solar cells on rough FTO substrates. *Energy Environ. Sci.* **18**, 3196-3210 (2025).
- [4] Lin, Y. et al. A Nd@C₈₂-polymer interface for efficient and stable perovskite solar cells. *Nature* **642**, 78-84 (2025).
- [5] He, D. et al. Universal ion migration suppression strategy based on supramolecular host-guest interaction for high-performance perovskite solar cells. *Adv. Mater.* **37**, 2505115 (2025).
- [6] Zhou, J. et al. Tailoring low-miller-index crystal facets realizes perovskite solar cells with flat grain-boundary grooves. *Angew. Chem. Int. Ed.* **64**, e202511042 (2025).
- [7] Peng, Y. et al. Heterogeneous-nucleation and epitaxial-growth of perovskite enabled by delafossite-type hole transport materials for high-performance solar cells. *Adv. Funct. Mater.* **36**, e21121 (2025).
- [8] Chen, X. et al. Entropy-driven strategy stabilizes photoactive halide perovskites for inverted solar cells. *Nat. Commun.* **16**, 9717 (2025).
- [9] Luo, C. et al. Dielectric molecular-bridges enable durable inverted perovskite solar cells with 26.60% efficiency and a high reverse breakdown voltage. *Energy Environ. Sci.* **19**, 338-348 (2026).
- [10] Khan, S.Y. et al. Revolutionizing the solar photovoltaic efficiency: A comprehensive review on the cutting-edge thermal management methods for advanced and conventional solar photovoltaics. *Energy Environ. Sci.* **18**, 1130-1175 (2025).

- [11] Park, S. et al. Synergistic enhancement of charge extraction and heat dissipation in inverted perovskite solar cells via N-doped top interlayers. *Energy Environ. Sci.* **17**, 8304-8312 (2024).
- [12] Li, J. et al. Facet-orientation-enhanced thermal transfer for temperature-insensitive and stable p-i-n perovskite solar cells. *eScience* **5**, 100372 (2025).
- [13] Wang, G. et al. Tuning thermal conductivity of hybrid perovskites through halide alloying. *Adv. Sci.* **11**, 2401194 (2024).
- [14] Tan, S. et al. Sustainable thermal regulation improves stability and efficiency in all-perovskite tandem solar cells. *Nat. Commun.* **15**, 4136 (2024).
- [15] Ma, Y. et al. High thermal conductivity of liquid crystal elastomer for stress-less flexible perovskite solar cells. *Adv. Funct. Mater.* **34**, 2405250 (2024).
- [16] Duda, J.C. et al. Thermal transport in organic semiconducting polymers. *Appl. Phys. Lett.* **102**, 251912 (2013).
- [17] Zhu, G. et al. Tuning thermal conductivity in molybdenum disulfide by electrochemical intercalation. *Nat. Commun.* **7**, 13211 (2016).
- [18] Balandin, A.A. Thermal properties of graphene and nanostructured carbon materials. *Nat. Mater.* **10**, 569-581 (2011).
- [19] Pogna, E.A.A. et al. Hot-carrier cooling in high-quality graphene is intrinsically limited by optical phonons. *ACS Nano* **15**, 11285-11295 (2021).
- [20] Wang, X. et al. Fluorocarbon-based solvent-bath annealing for high-performance perovskite photovoltaics. *Adv. Funct. Mater.* **35**, 2413400 (2025).
- [21] Ouyang, D. et al. Strategic synthesis of ultrasmall NiCo₂O₄ NPs as hole transport layer for

highly efficient perovskite solar cells. *Adv. Energy Mater.* **8**, 1702722 (2018).

[22] Aishwarya, K. et al. Carrier and morphologically tailored thermoelectric properties of MBi_2O_4 (M=Cu, Ni, and Zn) kusachiites. *Surf. Interfaces* **42**, 103440 (2023).

[23] Xiang, F. et al. Zigzag graphene nanoribbons with periodic porphyrin edge extensions. *Nat. Chem.* **17**, 1356-1363 (2025).

[24] Chen, S. et al. Thermal conductivity of isotopically modified graphene. *Nat. Mater.* **11**, 203-207 (2012).

[25] Chatterjee, S. et al. Introducing Cu_2O thin films as a hole-transport layer in efficient planar perovskite solar cell structures. *J. Phys. Chem. C* **120**, 1428-1437 (2016).

[26] Zhang, E. et al. Shape-controlled hydrothermal synthesis of CuFe_2O_4 nanocrystals for enhancing photocatalytic and photoelectrochemical performance. *Mater. Chem. Phys.* **235**, 121633 (2019).

[27] Muthukumaran, K.P. et al. Synthesis of spinel copper cobalt oxide for perovskite solar cells and electrocatalytic hydrogen evolution reaction. *J. Alloys Compd.* **1029**, 180789 (2025).

[28] Xu, W. et al. Efficient perovskite solar cells fabricated by Co partially substituted hybrid perovskite. *Adv. Energy Mater.* **8**, 1703178 (2018).

[29] Lee, J.H. et al. A solution-processed spinel CuCo_2O_4 as an effective hole transport layer for efficient perovskite solar cells with negligible hysteresis. *ACS Sustain. Chem. Eng.* **7**, 17661-17670 (2019).

[30] Sun, Q. et al. Critical review of Cu-based hole transport materials for perovskite solar cells: From theoretical insights to experimental validation. *Adv. Mater.* **36**, 2402412 (2024).

- [31] Liu, M. et al. Hierarchical nanostructures CuBi₂O₄ integrated with polyaniline nanofibrous for boosting hole extraction in carbon-based perovskite solar cells. *ACS Appl. Mater. Inter.* **16**, 56414-56426 (2024).
- [32] Chen, Y. et al. Design of an inorganic mesoporous hole-transporting layer for highly efficient and stable inverted perovskite solar cells. *Adv. Mater.* **30**, 1805660 (2018).
- [33] Wang, X. et al. Enhanced formaldehyde gas sensing performance of ternary CuBi₂O₄ oxides through oxygen vacancy manipulation and surface platinum decoration. *Sensor. Actuat. B-Chem.* **344**, 130190 (2021).
- [34] Chen, Y. et al. Thermally stable methylammonium-free inverted perovskite solar cells with Zn²⁺ doped CuGaO₂ as efficient mesoporous hole-transporting layer. *Nano Energy* **61**, 148-157 (2019).
- [35] Wu, Y. et al. Pushing the thermal conductivity limit by decoupling dual-channel phonon transport in crystals. *Nano Lett.* **26**, 1632-1638 (2026).
- [36] Ye, Q. et al. Abnormally low thermal conductivity of Co₂MnO₄ spinel induced by cation inversion. *J. Mater. Chem. A* **12**, 23761-23768 (2024).
- [37] Wang, Q. et al. Phonon dynamic behaviors induced by amorphous layers at heterointerfaces. *Phys. Chem. Chem. Phys.* **26**, 8397-8407 (2024).
- [38] Yin, Y. et al. Direct bonding of 6-inch GaAs/SiC wafers for enhanced thermal management in high-power devices. *ACS Appl. Mater. Inter.* **18**, 7527-7536 (2026).
- [39] Giri, A. et al. Interfacial defect vibrations enhance thermal transport in amorphous multilayers with ultrahigh thermal boundary conductance. *Adv. Mater.* **30**, 1804097 (2018).

- [40] Wang, M. et al. Anisotropic and ultralow phonon thermal transport in organic–inorganic hybrid perovskites: Atomistic insights into solar cell thermal management and thermoelectric energy conversion efficiency. *Adv. Funct. Mater.* **26**, 5297-5306 (2016).
- [41] Song, F. et al. Buried interface modification for high performance and stable inverted perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e202516012 (2025).
- [42] Huang, Y. et al. High-efficiency inverted perovskite solar cells via in situ passivation directed crystallization. *Adv. Mater.* **36**, 2408101 (2024).
- [43] Peng, W. et al. Reducing nonradiative recombination in perovskite solar cells with a porous insulator contact. *Science* **379**, 683-690 (2023).
- [44] Wang, S. et al. In situ self-elimination of defects via controlled perovskite crystallization dynamics for high-performance solar cells. *Adv. Mater.* **35**, 2305314 (2023).
- [45] Du, S. et al. Inhibiting perovskite decomposition by a creeper-inspired strategy enables efficient and stable perovskite solar cells. *Nat. Commun.* **15**, 5223 (2024).
- [46] Naveen, K. et al. Investigation of multiferroic behaviour at room temperature in Bi-induced orthoferrite: combined experimental and first principles studies. *B. Mater. Sci.* **43**, 196 (2020).
- [47] Gao, Y. et al. Rethinking the role of excess/residual lead iodide in perovskite solar cells. *Adv. Funct. Mater.* **33**, 2215171 (2023).
- [48] Jiang, Q. et al. Surface reaction for efficient and stable inverted perovskite solar cells. *Nature* **611**, 278-283 (2022).
- [49] Zhang, G. et al. Three birds with one stone: Triple defect passivation of tris(2,2,2-trifluoroethyl) phosphate enables 25.69% efficient inverted perovskite solar cells. *Angew. Chem.*

Int. Ed. **64**, e202502244 (2025).

[50] Peng, Y. et al. Enlarging moment and regulating orientation of buried interfacial dipole for efficient inverted perovskite solar cells. *Nat. Commun.* **16**, 1252 (2025).

[51] Zhang, H. et al. Multifunctional crosslinking-enabled strain-regulating crystallization for stable, efficient α -FAPbI₃-based perovskite solar cells. *Adv. Mater.* **33**, e2008487 (2021).

[52] Xu, Z. et al. Optimizing molecular packing and interfacial contact via halogenated N-glycidyl carbazole small molecules for low energy loss and highly efficient inverted perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e202503008 (2025).

[53] Yan, W. et al. Hot-carrier cooling regulation for mixed Sn-Pb perovskite solar cells. *Adv. Mater.* **36**, 2312170 (2024).

[54] Zheng, Y. et al. Towards 26% efficiency in inverted perovskite solar cells via interfacial flipped band bending and suppressed deep-level traps. *Energy Environ. Sci.* **17**, 1153-1162 (2024).

[55] Chen, Y. et al. Multilayer cascade charge transport layer for high-performance inverted mesoscopic all-inorganic and hybrid wide-bandgap perovskite solar cells. *Sol. RRL* **4**, 2000344 (2020).

[56] Yuan, R. et al. Boosted charge extraction of NbO_x-enveloped SnO₂ nanocrystals enables 24% efficient planar perovskite solar cells. *Energy Environ. Sci.* **14**, 5074-5083 (2021).

[57] Wang, Z.S. et al. Transient photovoltage measurements on perovskite solar cells with varied defect concentrations and inhomogeneous recombination rates. *Small Methods* **4**, 2000290 (2020).

[58] Yang, J. et al. Crystallization tailoring of cesium/formamidinium double-cation perovskite for efficient and highly stable solar cells. *J. Energy Chem.* **48**, 217-225 (2020).

[59] Li, Z. et al. Organometallic-functionalized interfaces for highly efficient inverted perovskite solar cells. *Science* **376**, 416-420 (2022).

[60] Wang, H. et al. Impurity-healing interface engineering for efficient perovskite submodules. *Nature* **634**, 1091-1095 (2024).

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Author Contributions

Yu C. conceived the idea and proposed the experimental design. X.W. and Yu C. performed the device fabrication and the technology innovation. X.W. and Yu C. characterized the material samples (AFM, SEM, in-situ UV-vis and MPP test). H.C., Y.P., J.Z. and S.L.L. assisted with the measurements. Y.S., E.D. and Y.W.D. helped perform the DFT and GIXRD measurements. Yu C. drafted the original manuscript. Yao C., Y.H.W., S.Y.P. and Q.P. helped with the manuscript

preparation. Yu C., S.Y.P. and Q.P. supervised the project. All the authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Figure Legends

Fig. 1| Preparation of CuBi_2O_4 -based composite HTLs. **a** The thermal conductivity of each layer. **b** XRD patterns of CuBi_2O_4 NCs. **c** Localized magnified HRTEM images of CuBi_2O_4 NCs. Scale bar represents 50 nm. **d** Lattice stripe of the (211) plane in CuBi_2O_4 (inset: lattice spacing of (211) planes). Scale bar represents 5 nm. **e** The thermal conductivity and the thermal diffusion coefficient of the CuBi_2O_4 substrate film. **f** The electronic and lattice thermal conductivity of CuBi_2O_4 calculated by DFT (inset: crystal structure diagram of CuBi_2O_4). **g** Schematic diagram of the construction of the DCZ-4P@ CuBi_2O_4 composite layer. **h** FT-IR spectroscopy of P=O stretching vibration for DCZ-4P and DCZ-4P@ CuBi_2O_4 respectively. **i** AFM-IR image for the DCZ-4P@ CuBi_2O_4 substrate. Scale bar represents 1 μm .

Fig. 2| Phonon-engineered heat conduction. **a** The thermal conductivity of different films at 300 K. **b-d** Phonon dispersion and phonon lifetime of the **b** DCZ-4P, **c** DCZ-4P@ NiO_x and **d** DCZ-4P@ CuBi_2O_4 models, respectively. **e-g** Group velocities for each phonon band as a function of frequency for the **e** DCZ-4P, **f** DCZ-4P@ NiO_x and **g** DCZ-4P@ CuBi_2O_4 samples, respectively. **h** The thermal-expansion coefficient of different films. **i** Simulations of heat transfer dynamics of different substrate films based on the finite element method. Scale bar represents 8 mm. **j** IR images of the control, NiO_x -based and CuBi_2O_4 -based perovskite films, respectively. Scale bar represents 2 cm.

Fig. 3| Perovskite crystallization dynamics and surface morphologies. **a-c** In-situ UV-vis absorption spectroscopy for the **a** control, **b** NiO_x-based and **c** CuBi₂O₄-based perovskite films, respectively. **d-f** In-situ PL spectroscopy for the **d** control, **e** NiO_x-based and **f** CuBi₂O₄-based perovskite films, respectively. **g** Top-view SEM images of different perovskite films. Scale bar represents 1 μm . **h** The linear fit $2\theta\text{-sin}^2\psi$ curves corresponding to the GIXRD spectra for the control, NiO_x-based and CuBi₂O₄-based films. **i** GIWAXS patterns of the control, NiO_x-based and CuBi₂O₄-based perovskite films. **j** The polar intensity profiles along the (100) ring at the range of 0.95~1.05 $\text{\AA}^{-1}$ for the control, NiO_x-based and CuBi₂O₄-based perovskite films.

Fig. 4| Carrier transport and recombination dynamics. **a-c** The pseudo-color TA plots (with a high-energy pump-light of 400 nm) of **a** glass/DCZ-4P/perovskite, **b** glass/DCZ-4P@NiO_x/perovskite and **c** glass/DCZ-4P@CuBi₂O₄/perovskite films, respectively. **d-f** Normalized pump-probe TA spectra of the **d** control, **e** NiO_x- and **f** CuBi₂O₄-based perovskite films at different delay time. **g** Hot-carrier temperature as a function of delay times for three films. **h** Energy loss rate as a function of the carrier temperature of perovskite samples. **i** Changes in resistance extracted from EIS spectra of three different devices.

Fig. 5| Photovoltaic performance of devices. **a** The champion J - V curves of the control, NiO_x-based and CuBi₂O₄-based inverted devices. **b** The statistical distribution curves of PCE for different devices. **c** The champion J - V curves of the control, NiO_x-based and CuBi₂O₄-based indoor devices. **d** Normalized PCEs versus time for unencapsulated PSCs under continuous thermal-aging at 85 °C in N₂-filled glove box. **e** Normalized PCEs versus time for encapsulated PSCs stored in 85% RH and 85 °C in the dark. **f** Normalized PCEs versus time for unencapsulated PSCs after thermal cycling in ambient air. **g** Dynamic MPP tracking of different devices under 1 sun illumination (white light LED).









